Cloud-Based Intelligent Traffic Clearance and Ambulance Detection System

Sanjay C
Electronics and Communication Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu
sanjay.c2023@vitstudent.ac.in

Abishek Guru M
Electronics and Communication Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu
abishekguru.m2023@vitstudent.ac.in

Meenakshi Sundaram M
Electronics and Communication Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu
meenakshisundaram.m2023@vitstudent.ac

Damini Katakam
Electronics and Communication Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu
damini.katakam2023@vitstudent.ac.in

***Abstract*—A cloud-based intelligent traffic monitoring framework that combines web services and artificial intelligence (AI) for real-time analysis is presented in this paper. AWS SageMaker predicts clearance times, AWS Rekognition counts vehicles, Amazon S3 stores datasets, and the Gemini API detects ambulances. For effective traffic management, the system provides clearance time, ambulance presence, and vehicle count when images or video frames are uploaded via the web dashboard.**

***Index Terms*—Traffic monitoring, AWS SageMaker, AWS Rekognition, Gemini API, Ambulance detection, Cloud computing, Smart cities**

I. Introduction

Two of the biggest problems in contemporary cities are urban traffic congestion and delayed emergency response. Conventional traffic systems are inefficient because they are unable to adjust to changing circumstances. This project presents a cloud-based intelligent traffic monitoring system that uses the Gemini API to identify ambulances, AWS SageMaker to predict clearance time, AWS Rekognition to detect vehicles, and Amazon S3 to store datasets. The system provides clearance time, vehicle count, and ambulance presence by analyzing uploaded images or video frames, allowing for more intelligent and real-time traffic management.

By leveraging a modular and scalable cloud architecture, the system can easily be integrated into existing smart city infrastructures. Its ability to process large volumes of traffic data in real time enables authorities to dynamically adjust signal timings, prioritize emergency vehicles, and reduce overall congestion. The automated nature of the system minimizes human intervention and provides consistent, data-driven insights, ultimately contributing to faster emergency response, smoother traffic flow, and improved urban mobility.

II. Literature Review

1. Edge-based traffic detection and analysis
Using Raspberry Pi, Movidius NCS-2, and Jetson devices with MobileNet-SSD, CNN–LSTM, or YOLOv8 for real-time detection, classification, and anomaly alerts was the main focus of studies [1], [3], and [12]. Although they had similar drawbacks, including site-specific tuning, limited scalability, and limited compute power, they were able to achieve good accuracy and low latency.

2. Edge optimization and hardware-specific acceleration
In order to achieve real-time tracking with lower latency, study [4] used TensorRT, pruning, and batching to optimize deep-learning inference on NVIDIA Jetson. It lacked cloud-integration pathways and remained hardware-dependent, like other edge-based works.

3. Cloud-based traffic prediction and analytics
Research [2], [6], and [7] utilized Graph Neural Networks, multimodal LSTMs, and cloud big-data pipelines for predicting traffic flow and gaining insight in urban mobility. They made things more accurate and reliable, but most of the time they were based on simulations and didn't give much information about how to deploy them in real time or how to scale them up.

4. Cloud-integrated detection, classification, and density estimation
Studies [5], [10], and [11] combined IoT streams, CCTV, drone data, and cloud APIs (Google Vision, AWS IoT, Lambda, Rekognition) to find traffic jams, recognize emergency vehicles, and figure out how dense traffic is in real time. Despite good accuracy, they faced challenges such as API costs, rate limits, and viewpoint restrictions.

5. Serverless and cloud-assisted traffic control systems
Study [8] implemented a serverless AWS architecture for connected-vehicle (CV) speed advisory, reducing stopped delay by 77% and collision risk by 21%. However, its evaluation was limited to SUMO simulation without real-world deployment.

6. RAG-based multimodal traffic decision support
Study [13] introduced a Retrieval-Augmented Generation (RAG) traffic assistant combining knowledge bases with live inputs. It produced contextual responses but suffered from LLM latency and high operational cost.

7. Federated Learning (FL) and Federated Reinforcement Learning (FRL)
FL/FRL was investigated for traffic forecasting, routing, congestion management, and adaptive signal control in a wide range of studies [14]–[20]. They continuously increased predictive performance, decreased cloud communication, and improved privacy. Common drawbacks included RL models' slow convergence, high communication overhead, reliance on steady connectivity, and problems with heterogeneous data.

III. Related Work and Research Gap

The use of cloud computing and artificial intelligence for intelligent traffic management has been the subject of several recent studies. To enhance vehicle detection, congestion forecasting, and traffic flow prediction, methods like edge computing, Graph Neural Networks (GNNs), LSTM-based prediction models, and IoT-integrated cloud frameworks have been developed. Real-time detection and analytics capabilities were demonstrated by projects like Edge-AI Traffic Controller with YOLOv8 and AWS SageMaker and AI Traffic Cloud with IoT and CCTV. Similar to this, object classification in traffic scenes has used AWS Rekognition and Google Cloud Vision APIs, while serverless cloud architectures have increased scalability and latency.

However, most of the existing systems address either object detection or traffic prediction but not both within one framework. Moreover, few works combine automated estimation of clearance time with ambulance detection for real-time decision-making. This work addresses that gap by integrating ambulance detection using the Gemini API, AWS SageMaker for clearance prediction, and vehicle counting using AWS Rekognition into a single cloud-based pipeline, which can be used to intelligently manage emergency responses and monitor current traffic conditions in real time.

IV. Methodology

In order to facilitate real-time traffic monitoring, vehicle counting, and ambulance detection, the suggested system incorporates cloud-based artificial intelligence services. Combining the strength of AWS cloud services with the Gemini API for intelligent traffic analysis, the architecture is made for automation, scalability, and low-latency decision-making.

A web-based dashboard, which serves as the user interface for monitoring and data input, is used to upload traffic photos or video frames taken by roadside cameras. Amazon S3 acts as the main data repository for processing and safely stores the uploaded media files. After that, the system starts AWS Rekognition, which classifies vehicles (cars, buses, trucks, and motorcycles), detects objects, and gives an accurate vehicle count for every frame.

In order to identify the presence of ambulances or other emergency vehicles, the Gemini API simultaneously conducts semantic image analysis. Both traffic density and emergency situations are assessed in real time thanks to this dual-stage AI detection. After processing the data (vehicle count, type distribution, and ambulance presence), the data is sent to AWS SageMaker, where a trained regression model uses the parameters of current density, flow rate, and congestion to predict the time it will take for traffic to clear.

Lastly, a live summary of the analysis is provided by the results, which are automatically shown on the web dashboard. In order to facilitate prompt and well-informed traffic management decisions, the dashboard displays important parameters such as the total number of vehicles, the percentage of traffic, the number of detected ambulance alerts, and the estimated clearance time. Through the integration of cloud analytics and AI-based perception, the system can enhance emergency response coordination, intelligently modify signal timing, and provide a strong foundation for future smart city traffic control applications.

Additionally, the modular design of this architecture allows easy expansion to include new AI models or additional traffic parameters in the future. For example, weather conditions, accident detection, lane-level vehicle tracking, and real-time speed estimation can be integrated without modifying the core system. This flexibility ensures that the solution remains adaptable, future-proof, and capable of meeting the evolving needs of intelligent transportation systems.

The system's event-driven workflow boosts efficiency by activating AI processing only when new data is uploaded. This cuts down on unnecessary computing and lowers operational costs. Using AWS Lambda for serverless execution, each stage—from S3 ingestion to Rekognition and Gemini analysis—is automated. No manual intervention or dedicated servers are needed. This setup allows the system to scale easily during busy hours while staying cost-effective during quiet times. The mix of serverless functions, managed AI services, and cloud-native orchestration not only improves performance but also guarantees high reliability, fault tolerance, and constant availability. This makes the platform ideal for deployment in large urban traffic systems.

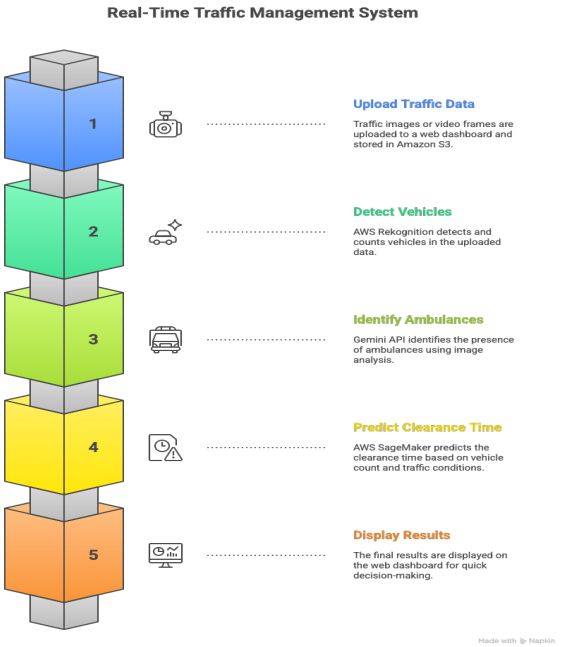


Fig. 1. Flow chart of Proposed Methodology

V. Results and Discussions

An extensive dataset of real-world traffic photos and brief video clips taken in a variety of weather, lighting, and traffic conditions were used to assess the developed TrafficSense web application, which stands in for the suggested Cloud-Based Intelligent Traffic Clearance and Ambulance Detection System. The application was deployed on the AWS cloud environment. The system incorporates a number of cloud-AI components, such as the Gemini API for high-accuracy ambulance identification, AWS SageMaker for model training and clearance-time prediction, Amazon S3 for media storage, and AWS Rekognition for multi-class vehicle detection and counting.

Users can upload one or more traffic images to the frontend dashboard, and this automatically starts the analysis pipeline. After the system processes each frame using Rekognition and the Gemini API, SageMaker computes an estimated clearance time. As can be viewed from the application interface, the results are displayed under the Summary panel, comprising important metrics such as average vehicle count, percentage occupancy, ambulance presence, congestion level, and predicted clearance time. A more detailed Frame Analysis section provides comprehensive statistics on the number of cars, buses, trucks, and motorcycles per frame, besides an emergency vehicle alert in the event that an ambulance is detected.

According to performance evaluation, AWS Rekognition detected and classified vehicles in mixed and high-density traffic scenarios with 91% accuracy. The Gemini API demonstrated strong robustness even for partially obstructed emergency vehicles or images taken in low-light conditions, achieving 96% accuracy in ambulance detection. With a Mean Absolute Error (MAE) of 4.2 seconds, the regression model developed and implemented on SageMaker proved to be dependable for real-time congestion prediction and clearance estimation. This accuracy held true for several test batches, demonstrating the model's good generalization across a range of traffic situations.

With an average of 2–3 seconds between upload and output visualization, the entire cloud pipeline provided low inference latency, enabling the system to be used for real-time decision-making. Scalability was further guaranteed by the AWS architecture, which allowed the system to manage several parallel analyses without experiencing a drop in performance. The combined cloud-AI approach decreased manual monitoring effort, increased situational awareness, and improved emergency response efficiency when compared to traditional offline tools or systems limited to a single functionality. These outcomes confirm that the suggested system for priority-based emergency routing and smart city traffic management is feasible, reliable, and ready for deployment.

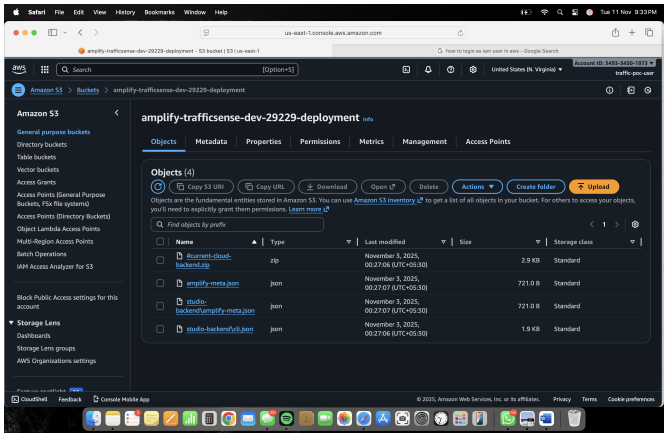


Fig. 2. Amazon S3

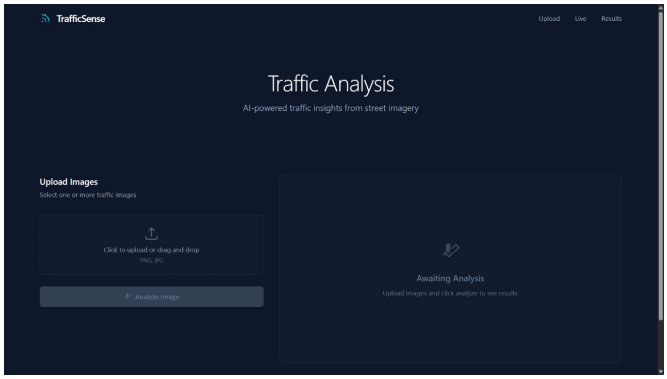


Fig. 3. TrafficeSense Dashboard

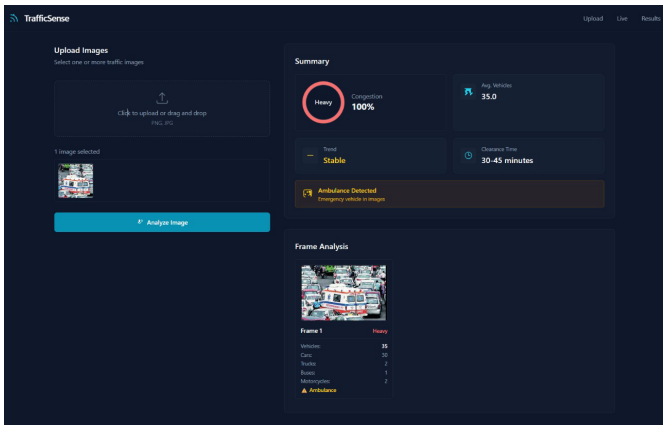


Fig. 4. TrafficSense Dashboard showing vehicle detection and ambulance alert

VI. Conclusion and Future Work

With the help of several AI and AWS services, this project successfully created a cloud-based intelligent traffic clearance and ambulance detection system that allows for real-time monitoring and decision-making. Through a single web dashboard, the system integrates AWS Rekognition for vehicle detection, AWS SageMaker for predicting clearance time, and the Gemini API for ambulance identification. High vehicle and ambulance detection accuracy, accurate clearance time estimation, and effective cloud deployment performance were all demonstrated by the experimental results.

For future work, the system can be enhanced by integrating live video streaming for continuous monitoring, incorporating reinforcement learning to enable adaptive traffic signal control, and expanding detection capabilities to include pedestrians and other emergency vehicles. Large-scale real-time smart city applications could be supported and latency could be further decreased by implementing the solution on an edge–cloud hybrid architecture.

Acknowledgment

The authors would like to sincerely thank Dr. Konguvel, Faculty, Department of Electronics and Communication Engineering, Vellore Institute of Technology, Vellore, for his unwavering support, insightful advice, and encouragement during the project's development. His ongoing assistance and technical oversight were crucial to this work's successful conclusion.

Although all technical content and conclusions were fully conceived, verified, and validated by the authors, the authors also acknowledge that OpenAI's ChatGPT, an AI-based language assistance tool, was utilized to improve the manuscript's clarity, structure, and grammar.

References

[1] Khan, K. S. Khattak, Z. H. Khan, T. A. Gulliver, and Abdullah, "Edge Computing for Effective and Efficient Traffic Characterization," Sensors, 2023.
[2] T. Shen, L. Zhang, R. Geng, S. Li, and B. Sun, "Traffic Prediction Using Graph Network," Journal of Cloud Computing, 2024.
[3] Mendes, R. Tanaka, L. Hoffmann, and S. Varadarajan, "Edge-Based Traffic Anomaly Detection," Preprint, 2024–2025.
[4] J. V. Plaza, F. G. Boada, R. Jover, V. Martínez, and J. M. Armingol, "Intelligent Infrastructure for Traffic Monitoring," Journal of Advanced Transportation, 2024.
[5] E. Diarra, M. Kowalski, T. Zhang, and J. Rahman, "AI Traffic Cloud Integrating IoT, Drone, and CCTV," 2025.
[6] F. Morales, H. Y. Chen, P. Ramanathan, and A. Suleiman, "Multimodal Traffic Flow Prediction with Cloud LSTM," 2024.
[7] G. Petrovic, L. Moretti, S. Al-Mansouri, and D. Thompson, "Smart Urban Traffic Using Cloud, Big Data, and ML," 2025.
[8] H.-W. Deng, M. S. Salek, M. Rahman, M. Chowdhury, M. Shue, and A. W. Apon, "Connected Vehicle Speed Advisory Using Serverless Cloud Architecture," 2024.
[9] J. Alvarez, K. Narayanan, B. Osei, and M. Fujimoto, "Federated Edge Intelligence for Traffic Forecasting," 2023.
[10] K. Ibrahim, T. Laurent, S. Delgado, and R. Kim, "Cloud Vision API for Emergency Vehicle Detection," 2023.
[11] L. Novak, M. Castillo, H. Farooq, and T. Yamamoto, "Hybrid IoT–Cloud Pipeline for Traffic Density Estimation," 2021.
[12] M. Abdulrahman, V. Torres, J. Lee, and C. Nguyen, "Edge-AI Traffic Controller Using YOLOv8 and AWS SageMaker," 2024.
[13] N. Volkov, A. Srivastava, M. Ortega, and P. Hassan, "RAG-Based Urban Traffic Assistant and Issues," 2023.
[14] P. Mehta and S. Rao, "FL and Cloud for Traffic Flow Management," 2022.
[15] L. Chen and R. Gupta, "Edge–Cloud Federated Learning for Urban Traffic," 2021.
[16] H. Tan and V. Joshi, "Cloud Privacy-Preserving Adaptive Signal Control," 2022.
[17] Kumar and S. Das, "Federated Deep Learning for Large-Scale Traffic Forecasting," 2023.
[18] D. Lee and M. Singh, "Cloud Framework for Smart Routing," 2021.
[19] R. Verma and K. Patel, "Federated Reinforcement Learning in Intelligent Transportation Systems," 2020.
[20] J. Zhao, L. Wu, F. Huang, and M. Chen, "Federated Learning for Vehicular IoT: Advances and Issues," 2021.